Using Blockchain to Track AI Model Origins

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Abstract

This paper investigates the design, implementation, and implications of leveraging blockchain technology to track the origins and lifecycle of artificial intelligence models. As AI systems proliferate across industries, provenance—the ability to verify the origin, training data, model modifications, and deployment history—becomes a critical property for trust, accountability, reproducibility, and regulatory compliance. This research articulates a comprehensive provenance framework that combines distributed ledger principles with metadata schemas, cryptographic signatures, secure off-chain storage, and governance rules to create verifiable, tamper-evident trails for model artifacts. The study develops a prototype architecture that demonstrates how model weights, training datasets, training configurations, and update events can be recorded in an auditable manner without exposing sensitive data. The prototype balances on-chain immutability with off-chain scalability and privacy controls by employing hashed pointers and selective disclosure mechanisms. The paper reports on an experimental evaluation that measures performance overhead, storage costs, and the fidelity of provenance records under realistic development workflows. It also discusses the legal, ethical, and organizational implications of blockchain-backed provenance, outlining the trade-offs between transparency, privacy, and operational efficiency. The findings indicate that blockchain-enabled provenance can substantially enhance model accountability in high-stakes settings while requiring thoughtful integration with data governance, cryptography, and organizational processes. The research concludes with an agenda for future work, including standardization of provenance metadata, integration with model supply chains, and policy measures to incentivize adoption.

Keywords: blockchain, provenance, AI model provenance, model governance, distributed ledger, accountability, metadata, cryptographic signatures

Introduction

The rapid adoption of artificial intelligence across healthcare, finance, public administration, and critical infrastructure has produced tremendous value while introducing significant concerns surrounding trust, safety, and accountability. One central challenge is provenance: determining where an AI model came from, how and on what data it was trained, which pre-processing steps were used, which architectures and hyperparameters were employed, who contributed to its development, and how it has been modified over time. Provenance matters for reproducibility, liability allocation, regulatory compliance, and the detection of bias or malfeasance. When models perform poorly, cause harm, or are implicated in legal disputes, robust provenance records can

reveal whether errors derive from flawed data, improper training, malicious tampering, or misconfiguration.

Current practices for managing model provenance are fragmented. Teams maintain internal logs, version control systems, and ad hoc metadata stores. These systems are often centralized, siloed, and susceptible to tampering or loss. Additionally, provenance information can be incomplete or inconsistent, especially when models pass through complex supply chains involving third-party components, pre-trained modules, or outsourced training services. The absence of tamper-evident, interoperable provenance mechanisms undermines trust and complicates accountability.

Blockchain technology, with its properties of decentralization, immutability, and cryptographic verifiability, offers a promising substrate for recording provenance. A distributed ledger can provide an append-only, tamper-evident record of model lifecycle events such as creation, training, evaluation, deployment, and update. When combined with cryptographic signing, access control, and off-chain storage for bulky artifacts, blockchains can enable stakeholders to verify that a model claiming a particular origin genuinely traces back to a documented history. Moreover, programmable smart contracts can encode governance rules, consent mechanisms, and automated verification processes that facilitate compliance and redress.

This paper explores the potential and limitations of using blockchain to track AI model origins. The contribution is threefold. First, we propose a provenance framework tailored to AI models that reconciles blockchain's strengths with the privacy and scalability requirements of machine learning workflows. Second, we detail a prototype implementation and experimental evaluation that measure the cost, latency, and accuracy of the approach under realistic scenarios. Third, we analyze the broader ethical, legal, and organizational implications, discussing how provenance can be operationalized across model supply chains and what governance structures are necessary to balance transparency and confidentiality.

Literature Review

The ecosystem of AI development comprises numerous artifacts and processes that must be tracked for meaningful provenance. The primary artifacts include raw datasets, data processing scripts, annotated data labels, feature extractors, model architectures, model weights, training hyperparameters, evaluation metrics, validation datasets, and deployment manifests. Each of these artifacts may originate from different entities and may be subject to confidentiality constraints. Provenance for AI therefore requires mechanisms for fine-grained metadata capture, cryptographic binding of artifacts to records, and flexible access control.

Provenance concepts have long been studied in the contexts of databases, scientific workflows, and digital forensics. Core principles include capturing provenance at the right level of granularity, ensuring non-repudiation of records, enabling chain-of-custody tracking, and supporting queries that can reconstruct the history of an artifact. Translating these principles to AI necessitates handling large binary artifacts like model weights and training datasets. Efficient provenance must therefore combine on-chain references with secure off-chain stores to maintain fidelity without overwhelming blockchain capacity.

Blockchain introduces unique advantages for provenance. Its append-only ledger provides an auditable timeline that is resistant to single-point tampering. Cryptographic primitives allow signatures and hashes to serve as immutable fingerprints for digital artifacts. Smart contracts introduce programmability enabling automatic enforcement of governance policies. Nonetheless, blockchain presents trade-offs. Storing large files on-chain is impractical due to cost and scalability constraints. Public ledgers raise privacy concerns, as provenance data might reveal sensitive information. Permissioned ledgers mitigate exposure but involve trust trade-offs and governance complexity. The literature has proposed hybrid architectures that place fingerprinting and metadata on-chain while preserving actual artifacts in secure off-chain repositories.

Several technical approaches have emerged to make blockchain-based provenance practical. Hash-based commit models calculate digests of artifacts and store those digests on the ledger, providing integrity verification while keeping data off-chain. Merkle trees and Merkle DAG structures allow aggregation of many artifacts into a single succinct root that can be recorded in a transaction. Content-addressable storage and persistent storage networks allow mutable access to off-chain artifacts while preserving immutability through cryptographic referencing. Zero-knowledge proofs and selective disclosure techniques enable verification of assertions about private data without revealing the data itself. Interoperability layers that map provenance metadata to common schemas and ontologies are essential for cross-organization verification, and standardization efforts have begun to define provenance vocabularies and exchange formats.

Governance, incentives, and legal considerations are central in the literature. Provenance systems must align incentives across developers, vendors, regulators, and end users. Without adequate incentives, actors may avoid recording provenance or may manipulate records off-chain. Legal frameworks influence the design of provenance mechanisms, especially when they intersect with data protection requirements, intellectual property, and contractual liabilities. There is growing emphasis on embedding governance logic in smart contracts to manage permissions, attestations, and dispute resolution, but smart contracts also raise new challenges related to formal verification and upgradeability.

Empirical evaluations within industry settings reveal both the promise and the friction points of blockchain-enabled provenance for models. Prototype deployments show that provenance can deter malicious behavior, improve reproducibility, and simplify regulatory audits. However, concerns persist regarding the additional operational overhead, the need for robust key management, and the complexity of integrating heterogeneous tooling chains. The literature stresses the need for pragmatic, hybrid architectures that combine the strengths of blockchain with established software engineering and data governance practices.

From a socio-technical standpoint, provenance is not purely a technical problem. Organizational processes, norms, and legal contracts shape how provenance is captured, validated, and used. There's a need for standards, external certification, and interoperable tooling that lower the barrier to adoption. Provenance systems should be designed to support diverse stakeholder goals, including transparency for affected individuals, accountability for organizations, and flexibility for developers.

Methodology

This research adopts a design-science methodology combining system design, prototype implementation, and experimental evaluation. The methodology centers on defining functional requirements, designing a hybrid blockchain-off-chain architecture, implementing a prototype, and evaluating the prototype against performance and usability metrics. The study also incorporates qualitative analysis of governance and policy implications.

Functional Requirements

The provenance system must satisfy the following requirements: verify the authenticity and integrity of model artifacts; record lifecycle events in an immutable and auditable manner; protect sensitive data while enabling verifiable claims; support selective disclosure of provenance information; be interoperable across organizations and tooling; and impose acceptable operational overhead. These requirements reflect practical constraints encountered in modern machine learning pipelines.

System Design

The design follows a hybrid approach. Core provenance metadata and cryptographic fingerprints are recorded on-chain to leverage immutability and decentralization. Bulky artifacts such as datasets and model weight files remain off-chain in secure storage systems. On-chain records contain structured metadata fields, digests, and pointers to off-chain content. Smart contracts implement registries for models, attestations, and access control policies. A layer of identity and key management ensures that provenance entries are cryptographically signed by the responsible entities.

Key components include an Artifact Manager that prepares artifacts for registration, a Storage Layer which manages off-chain storage and retrieval, a Blockchain Registry that records provenance events, a Verification Engine that validates on-chain records against off-chain content, and a Governance Module that enforces access and attestation policies. The system supports event types such as create, train, evaluate, deploy, update, fork, and deprecate. Each event includes a timestamp, actor identity, cryptographic signature, pointers to artifacts, and optional declarative assertions about compliance or certification.

Privacy and Selective Disclosure

To protect sensitive information, the design uses hashed pointers and redactable metadata strategies. Sensitive fields are hashed prior to being published on-chain. Selective disclosure is enabled through cryptographic techniques: actors can reveal preimage data to verifiers or generate zero-knowledge proofs that a particular property holds without disclosing the underlying data. The governance module governs who can request disclosure and under which conditions.

Prototype Implementation

We implemented a prototype that integrates an enterprise permissioned blockchain with a content-addressable off-chain storage system. The blockchain manages a registry smart contract that

records provenance records. The off-chain store supports secure storage and retrieval of model artifacts and generates content digests that are bound to on-chain records.

The prototype simulates typical model lifecycle workflows where a dataset owner uploads data, a training organization conducts training with specified hyperparameters, and an auditor verifies the provenance. Key management uses an enterprise PKI where actors hold signing keys managed by a hardware security module. The Artifact Manager computes SHA-256 digests for artifacts and constructs Merkle roots for composite artifacts. The smart contract API accepts a compact provenance record that includes the Merkle root, actor signature, timestamp, and a pointer to the off-chain manifest.

Evaluation Setup

The evaluation aims to measure provenance fidelity, performance overhead, storage costs, and verification latency. Test scenarios include model creation and registration, model update with multiple incremental training runs, and verification by a third-party auditor. Datasets of varying sizes and model artifacts of varying complexity were used. Metrics collected include on-chain transaction latency, gas or ledger cost equivalents, off-chain storage access time, and time to complete end-to-end verification. We also measured the size of on-chain records under different metadata granularities to assess cost scalability.

Security and Threat Modeling

We conducted a threat analysis to understand potential adversarial behaviors, including attempts to tamper with off-chain artifacts, misrepresenting provenance entries by colluding actors, key compromise, and denial-of-service attacks against the blockchain registry. The design incorporates countermeasures such as multi-signature attestations, time-stamping from multiple independent witnesses, and challenge-response protocols for off-chain artifact verification.

Ethical and Legal Review

Given the privacy implications of provenance, the research included an ethical and legal review of the prototype. Particular attention was paid to compliance with privacy regulations, ownership claims, and the risk of exposing sensitive training data through provenance metadata. The system was designed to be configurable for different legal regimes, enabling stronger privacy controls where required.

Results

The prototype demonstrated that blockchain-backed provenance can record and verify complex model lifecycles while balancing privacy and scalability. The evaluation produced results across technical metrics and qualitative considerations.

Provenance Fidelity

The system successfully captured lifecycle events with sufficient detail to reconstruct model histories. The combination of Merkle-rooted artifact digests and signed metadata enabled verification workflows to confirm that a given model artifact corresponded to the recorded provenance entry. The Merkle approach allowed aggregates of dataset versions, preprocessing scripts, and model checkpoints to be represented compactly on-chain, enabling auditors to verify entire composite artifacts by checking Merkle proofs against off-chain content.

Performance Overhead

On-chain recording introduced latency attributable to transaction processing. In the permissioned blockchain environment used for the prototype, median transaction latency for writing a provenance record was modest and compatible with development workflows. The cost of writing metadata scaled with the size of the metadata payload. Keeping the on-chain payload minimal through hashing and succinct metadata substantially reduced the fee-like cost in our permissioned setting. Off-chain storage retrieval and verification dominated overall verification latency when large datasets or model checkpoints were involved. However, the verification process could be parallelized and amortized across audits to achieve practical performance.

Storage Costs

Storing only digests and compact manifests on-chain proved cost-effective. The off-chain storage bore the bulk of storage expense, and content-addressability facilitated deduplication, which reduced storage duplication across multiple model versions. The use of Merkle trees further minimized the need to duplicate data references, because shared subtrees could be referenced across model versions.

Scalability

The hybrid architecture scaled linearly with the number of provenance events in terms of transaction count. Bulk registration strategies, such as batching multiple small events into a single atomic record using aggregated Merkle roots, reduced the number of transactions required and improved throughput. The trade-off is between granularity of on-chain records and transaction volume; coarser granularity reduces transaction loads but can make fine-grained audit queries more complex. The system supports both approaches, allowing organizations to choose the balance that fits their operational model.

Security Analysis

The threat analysis demonstrated that cryptographic binding of artifacts to ledger entries prevented straightforward tampering. If an adversary modified off-chain content without updating the ledger, verification would detect the inconsistency because the computed digest would not match the on-chain record. Multi-signature attestations and independent witness time-stamps increased resistance to collusion scenarios where a single actor attempts to assert false provenance. Key compromise remains a critical vulnerability; mitigation requires robust key management, hardware-based key storage, and revocation mechanisms. The prototype supported certificate revocation lists and key rotation policies to address this risk.

Privacy and Disclosure

Selective disclosure mechanisms worked effectively in scenarios where auditors needed evidence without full data access. Hashing sensitive metadata on-chain and revealing preimages only upon authorized requests provided a workable trade-off between transparency and confidentiality. The prototype also experimented with zero-knowledge proof constructs for simple properties, such as proving that a dataset contained at least a given number of samples without revealing sample-level details. These primitives are promising but require careful engineering to scale to complex assertions.

Usability

Integrating provenance recording into typical machine learning workflows was feasible with minimal friction when supported by tooling that automated artifact hashing, manifest creation, and transaction submission. Developers appreciated that provenance recording could be invoked as part of CI/CD pipelines. However, the necessity of signing transactions and managing keys introduced some operational complexity that organizations must address through training and tooling.

Interoperability

The prototype demonstrated that a standardized metadata schema for provenance facilitated cross-organization verification. When participants adhered to a common manifest structure, auditors could reliably parse and verify records. The lack of universal standards remains an adoption barrier, and the prototype emphasized the need for industry-wide metadata vocabularies that support both technical and legal queries.

Discussion

The findings indicate that blockchain-enabled provenance offers a compelling approach to enhancing the traceability and accountability of AI models. Yet, practical adoption depends on reconciling several tensions: transparency versus privacy, immutability versus the need for lawful deletion or redaction, and decentralization versus governance.

Transparency and Privacy

Transparency is valuable for trust, reproducibility, and auditability. Immutable records provide a reliable chain of custody that can deter malfeasance. However, transparency can conflict with privacy obligations and intellectual property protection. Recording raw datasets or detailed preprocessing steps on a public ledger is unacceptable in many contexts. The hybrid approach presented addresses this by keeping sensitive data off-chain and recording cryptographic fingerprints on-chain. Selective disclosure and privacy-preserving proofs allow verifiers to obtain necessary assurances without exposing raw data. Nonetheless, implementing selective disclosure adds complexity and requires standardized policies to determine when and how disclosures occur.

Immutability and Redaction

Blockchain immutability is an asset for provenance but can be problematic when legal frameworks demand data erasure or when mis-recorded metadata must be corrected. One pragmatic approach is to record pointers to mutable off-chain manifests and to record redaction or deprecation events on-chain. In this pattern, the ledger preserves a complete audit trail, including redaction events, while allowing the underlying content to be managed off-chain in accordance with legal obligations. Legal frameworks and contractual clauses must account for this architecture to ensure that immutability does not inadvertently contravene regulatory requirements.

Decentralization and Governance

The choice between permissioned and permissionless ledgers has significant implications. Permissionless systems provide greater decentralization but raise privacy and performance challenges. Permissioned ledgers allow governance and access control suitable for enterprise settings but require agreement on governance rules and trusted participants. Many practical deployments will favor permissioned architectures within consortiums or industry bodies where participants require oversight and privacy. Regardless of ledger type, robust governance frameworks are necessary to define roles, responsibilities, dispute resolution mechanisms, and standards for metadata and attestation.

Operational Overhead and Incentives

Provenance systems introduce operational overhead, including transaction management, key lifecycle management, and off-chain storage costs. These overheads are manageable for entities with significant regulatory exposure or reputational risk, but they may be prohibitive for smaller actors without clear incentives. Industry consortia and regulatory encouragement can help align incentives by making provenance a requirement or a competitive differentiator. Tooling that automates provenance capture and integrates with existing development pipelines reduces friction and increases adoption likelihood.

Legal and Ethical Considerations

Provenance can play a crucial role in assigning liability and demonstrating compliance. However, provenance records are pieces of evidence that must be trustworthy and legally admissible. Ensuring legal robustness requires careful design of signing practices, time-stamping mechanisms, and chain-of-custody documentation, as well as alignment with evidentiary standards. Ethically, provenance must be designed to protect affected individuals, especially when provenance metadata could reveal sensitive attributes about training data contributors. Privacy-preserving techniques and governance policies must safeguard these interests.

Standardization and Interoperability

A central barrier to widespread adoption is the lack of common standards for model provenance metadata. The prototype underscored the importance of schema alignment across organizations. Standardization efforts should define minimum provenance fields that capture essential lifecycle events and support the queries needed by auditors, regulators, and end users. Interoperability also

requires common cryptographic practices and identity frameworks so that attestations from different entities are meaningful and verifiable.

Future Directions

Several research directions deserve priority. First, richer privacy-preserving verification techniques should be developed and optimized for practical properties relevant to model provenance. Second, integration of provenance with model governance workflows and supply chains—such as marketplaces for pre-trained models—should be explored to handle trust in outsourced components. Third, empirical studies of governance models and incentive structures can illuminate which approaches produce robust, widely adopted provenance ecosystems. Finally, there is a need for legal and policy scholarship to clarify how blockchain provenance interacts with data protection, intellectual property, and liability regimes.

Conclusion

This research demonstrates that blockchain-backed provenance can materially improve the traceability of AI models by providing cryptographically verifiable, tamper-evident records of lifecycle events. The hybrid architecture that records compact cryptographic fingerprints on-chain while storing artifacts off-chain strikes an effective balance between immutability and scalability. Operational realities, privacy concerns, and governance complexities require careful engineering, robust key management, and collaborative standardization to make provenance practical across industry settings. When implemented thoughtfully, blockchain-enabled provenance has the potential to strengthen trust, facilitate audits, and provide a foundation for responsible AI deployment.

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